

The correlation between mid-latitude trough and the plasmapause

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Abstract

[1] We use simultaneous global observations of the mid-latitude trough and the plasmapause to experimentally prove a long-standing conjecture of magnetosphere-ionosphere coupling—namely the mid-latitude trough and plasmapause are on the same field line. Global Ionospheric Maps (GIM), generated using ground based GPS receivers, are used to detect the globally extended mid-latitude trough; while global IMAGE EUV pictures are used to estimate the plasmapause position. Observations during the equinox and solstices and during quiet and disturbed periods are analyzed. In addition, positions of the mid-latitude trough are calculated using a simple empirical model. The two independent observations (mid-latitude trough and plasmapause positions) and an empirical model have been compared on a global scale and found to be in excellent agreement.

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